David Leo Barack, Ph.D.

Curriculum Vitae

https://orcid.org/0000-0001-7131-4638

Postdoctoral Research Fellow
Department of Philosophy
Department of Social and Decision Sciences
Carnegie Mellon University
Resident Fellow
Center for Philosophy of Science
University of Pittsburgh
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Employment

2026 –	Postdoctoral Research Fellow, Carnegie Mellon University, Department of Philosophy and Department of Social and Decision Sciences; Resident Fellow, University of Pittsburgh, Center for Philosophy of Science
2024 – 2025	Research Assistant Professor, Lingnan University, Department of Philosophy
2024	Postdoctoral Fellow, University of Antwerp, Department of Philosophy
2020 – 2023	Research Associate, University of Pennsylvania, Platt Lab, Department of Neuroscience and Visiting Scholar, Department of Philosophy
2015 – 2020	Presidential Scholar in Society and Neuroscience and Associate Research Scientist, Columbia University, Zuckerman Mind, Brain, and Behavior Institute and Center for Science and Society, Departments of Philosophy, Neuroscience, and Economics

Education

2008 – 2014	Duke University Ph.D. Philosophy; Dissertation: "Mindcraft: A Dynamical Systems Theory of Cognition"; Committee: Alex Rosenberg (Co-Chair), Walter Sinnott-Armstrong (Co-Chair), Felipe De Brigard, Karen Neander, Michael Platt Cert. Cognitive Neuroscience
2007 – 2008	Northwestern University Cert. Pre-Medical (unfinished)
2005 – 2007	University of Wisconsin-Milwaukee

	M.A. Philosophy; Thesis: “Systematicity and Mental Content”; Committee: Robert Schwartz (Chair), John Koethe, Steven Leeds
2000 – 2004	Pitzer College B.A. Consciousness Studies; Thesis: “Regaining Consciousness: An exploration of consciousness, the mind, processes and their scientific study”; Advisors: William Banks (Pomona College), Amy Kind (Claremont McKenna College), Brian Keeley (Pitzer College)

Areas of Specialization

Neuroscience

Neural mechanisms of decision making and foraging behavior

Philosophy

Philosophy of science (philosophy of neuroscience; philosophy of cognitive science) and philosophy of mind

Areas of Competence

Neuroscience

Behavioral decision-making and neural mechanisms of reward processing, computational modeling of behavior

Philosophy

Logic, Metaphysics, philosophy of AI

Publications

Philosophy

Books

Edited:

1. Rowbottom DP, Curtis-Trudel A, and Barack DL. The Role of Artificial Intelligence in Science: Methodological and Epistemological Studies. Forthcoming, Routledge.

Under Review:

1. Barack DL. A Symphony for Thought. Oxford University Press.

Journal Articles

Reviewed:

1. Pohl S, Walker EY, Barack DL, Lee J, Denison RN, Block N, Meyniel F, and Ma WJ. “Desiderata of evidence for representation in neuroscience”. 2026. *Nature Reviews Neuroscience*, v27, p. 357-372. https://www.nature.com/articles/s41583-026-01030-8
2. Barack DL. “The Multiple Action Account of Filling a Role”. 2025. *Philosophical Studies*, v182(8), p. 2297-2312.

3. Barack DL. “Theory change in cognitive neurobiology: the case of the orbitofrontal cortex”. 2025. *Wiley Interdisciplinary Reviews: Cognitive Science*, 16:e70003, p. 1-12.
4. Barack DL. “Context and Neural Function”. 2025. *Philosophy of Science*, v92, p. 185-203.
5. Barack DL. “Rational Dynamics in Efficient Inquiry”. 2025. *Analysis*, v85(3), p. 649-658.
6. Barack DL. “What is Foraging?” 2024. *Biology and Philosophy*, v39. https://link.springer.com/article/10.1007/s10539-024-09939-z
7. Barack DL. “The Dynamicist Landscape”. *Topics in Cognitive Science*. 2023. https://onlinelibrary.wiley.com/doi/epdf/10.1111/tops.12699?domain=author&token=BJGWWYENFMBRSMYBMWJS
8. Barack DL and Krakauer J. “Two Views on the Cognitive Brain”. 2021. *Nature Reviews Neuroscience*. https://www.nature.com/articles/s41583-021-00448-6
9. Barack DL. “Mental Kinematics”. 2021. *Synthese*, v199, p. 1091-1123. https://link.springer.com/article/10.1007/s11229-020-02766-1
10. Barack DL and Jaegle A. “Codes, Functions, and Causes”. 2019. *Behavioral and Brain Sciences*. https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/abs/codes-functions-and-causes-a-critique-of-brettes-conceptual-analysis-of-coding/917AFE6FA4850F278AE8D93E28B6BB92
11. Barack DL. “Mental Machines”. 2019. *Biology and Philosophy*, v34(63). https://link.springer.com/article/10.1007/s10539-019-9719-6
12. Barack DL. “Cognitive Recycling”. 2019. *British Journal for the Philosophy of Science*, v70(1), p. 239–268. https://www.journals.uchicago.edu/doi/10.1093/bjps/axx024

Invited:

1. Flanagan OJ and Barack DL. “Neuroexistentialism”. 2010. *EurAmerica*, v40(3), p. 1–18.

Revise and Resubmit:

1. Barack DL. “Boundaries of Cognition”. *Mind and Language*.
2. Barack DL. “Understanding and Neural Networks”. *Synthese*.

Under Review:

1. Barack DL. “Scientific Understanding and Evidence”. *Philosophy and Phenomenological Research*.

Manuscripts:

1. Barack DL and Jaegle A. “Extended Intelligence”. arXiv: https://arxiv.org/abs/2209.07449.

Chapters

1. Barack DL. “Perils and Prospects of NeuroAI”. In Rowbottom et al. ed. Forthcoming.

Book Reviews

1. Barack DL. “The Tools of Neuroscience Experiment by John Bickle, Carl F Craver and Ann-Sophie Barwich”. 2022. *British Journal for the Philosophy of Science.*

Neuroscience

Journal Articles

Reviewed:

1. Barack DL, Ludwig VU, Parodi F, Ahmed N, Brannon EM, Ramakrishnan A, and Platt ML. “Attention Deficits Linked with Proclivity to Explore while Foraging”. 2024. *Proceedings of the Royal Society B*, v291(2017), p. 1-13. https://royalsocietypublishing.org/doi/10.1098/rspb.2022.2584
2. Walker EY, Pohl S, Denison RN, Barack DL, Lee J, Block N, Ma WJ, and Meyniel F. “Studying the Neural Representations of Uncertainty”. 2023. *Nature Neuroscience*. https://doi.org/10.1038/s41593-023-01444-y
3. Kay K, Bonnen K, Denison RN, Arcaro MJ, and Barack DL. “Tasks and their role in visual neuroscience”. 2023. *Neuron*. https://www.cell.com/neuron/fulltext/S0896-6273(23)00218-0
4. Barack DL, Bakkour A, Shohamy D, and Salzman CD. “Signatures of visuospatial information foraging during learning of complex environments”. 2023. *Scientific Reports*. https://www.nature.com/articles/s41598-023-27662-9.pdf
5. Barack DL, Miller EK, Moore CI, Packer AM, Pessoa L, Ross LN, and Rust NC. “A call for more clarity around causality in neuroscience”. 2022. *Trends in Neurosciences*. https://www.cell.com/trends/neurosciences/fulltext/S0166-2236(22)00121-7
6. Barack DL and Krakauer J. “Two Views on the Cognitive Brain”. 2021. *Nature Reviews Neuroscience*. https://www.nature.com/articles/s41583-021-00448-6
7. Barack DL and Platt ML. “Neuronal activity in the posterior cingulate cortex signals environmental information and predicts behavioral variability during routine foraging”. 2021. *Journal of Neuroscience*, v41(12), p. 2703-2712. https://www.jneurosci.org/content/41/12/2703.abstract
8. Barack DL, Chang SWC, and Platt ML. “Posterior cingulate neurons dynamically signal decisions to disengage during foraging”. 2017. *Neuron*, v96(2), p. 339-347. https://www.sciencedirect.com/science/article/pii/S0896627317309170
9. Addicott MA, Pearson JM, Sweitzer MM, Barack DL, and Platt ML. “A primer on foraging and the explore/exploit trade-off for psychiatry research”. 2017. *Neuropsychopharmacology*, v42, p. 1931-1939. https://www.nature.com/articles/npp2017108
10. Barack DL and Platt ML. “Neurocomputational Nosology: Malfunctions of Models and Mechanisms”. 2016. *Frontiers in Psychology*, v7(602), p. 1–9. https://www.frontiersin.org/articles/10.3389/fpsyg.2016.00602/full
11. Barack DL and Gold JI. “Temporal Trade-offs in Psychophysics”. 2016. *Current Opinion in Neurobiology*, v37, p.121–125. https://www.sciencedirect.com/science/article/pii/S0959438816300010

12. Chang SWC, Barack DL, and Platt ML. "Mechanistic Classification of Neural Circuit Dysfunctions: Insights from Neuroeconomics Research in Animals". 2012. *Biological Psychiatry*, v72, p. 101-106. https://www.sciencedirect.com/science/article/pii/S0006322312001448
13. Pearson JM, Heilbronner SR, Barack DL, Hayden BY, and Platt ML. "Posterior Cingulate Cortex: Adapting Behavior to a Changing World". 2011. *Trends in Cognitive Sciences*, v15(4), p. 143–151. https://www.sciencedirect.com/science/article/pii/S1364661311000180

Invited:

1. Wise TB, Barack DL, and Templer VL. "Geometrical representation of serial order in working memory". 2022. *Learning and Behavior*. https://doi.org/10.3758/s13420-022-00541-5

Under Review:

1. Clarkson T, Arora A, Quarmley M, Barack DL, Johnston C, Carpenter A, Archambault C, Kendal P, Hackel L, and Jarcho JM. "Learning to Fear Social Interactions: Dysregulated Neural Mechanisms of Social Learning in Adolescent Social Anxiety". *Biological Psychiatry*.
2. Hedden, A, Barack DL*, and Bakkour A*. "Costly Curiosity in Older Adults". *Proceedings of the National Academy of Sciences*. *: co-senior authors and equal contributions.

Revise and Resubmit:

1. Barack DL, Parodi F, Ludwig V, and Platt ML. "Information gathering explains decision dynamics during human and monkey reward foraging". *PNAS*. bioRxiv: https://www.biorxiv.org/content/10.1101/2023.10.14.562362v1

Chapters

1. Barack DL and Platt ML. "Engaging and Exploring: Cortical Circuits for Adaptive Foraging Decisions". 2017. In J.R. Stevens (ed.), Impulsivity, p. 163-199. https://link.springer.com/chapter/10.1007/978-3-319-51721-6_6
2. Harris JA, Barack DL, McMahon AR, Mitroff SR, & Woldorff MG. "Object-Category Processing, Perceptual Awareness, and the Role of Attention during Motion-Induced Blindness". 2014. In G. R. Mangun (ed.), Cognitive Electrophysiology of Attention: Signals of the Mind, p. 97–106.

Grants

National Institutes of Health/NIDA – 1K99DA048748-01 – (05/2019 – 05/2023)
K99/R00 Pathway to Independence BRAIN Initiative Diversity Award
Role: Principal Investigator
$318,104 (K99 phase); $750,000 (R00 phase)

National Institutes of Health – T32 Training Fellowship
Administered by Columbia University
Role: Post-doctoral Fellow
$52,000

Presidential Scholarship in Society and Neuroscience
Columbia University
Role: Principal Investigator
$325,000

Summer research stipend
Duke University
Role: Graduate student
$2000

Talks

Symposia

"Reconciling the Dichotomy between Sherringtonian and Hopfieldian Views on Neural Computations". Generative Adversarial Collaboration, Cognitive Computational Neuroscience 7, August 2023, https://www.youtube.com/watch?v=cl_Wa7CGm7A.

"Content Preservation and Streams of Thought". American Philosophical Association, Eastern Division, January 2023.

Philosophy

"Transitions in Mind". American Philosophical Association, Pacific Division, April 2027.

TBD.* American Philosophical Association, Eastern Division, January 2027.

"Zetetic Agency".† American Philosophical Association, Eastern Division, January 2027.

"Norms of Pursuit for Scientific Investigation". Philosophy of Science Association, November 2026.

"Scientific Understanding and Evidence".* University of Pittsburgh, November 2026.

"Neural Doctrines". Philosophy of Neuroscience in the Gulf IX, September 2026.

"Zetetic Agency".* University of Turku, August 2026.

"Plans, Planning, and Intentions". European Society for Philosophy and Psychology, July 2026.

"Zetetic Agency".* Tulane University, April 2026.

"Zetetic Agency".* Universitat Pompeu Fabra, March 2026.

"Neural Dynamics: Real, Latent Features of Neural Activity".* Stanford University, Workshop on Neural Dynamics, March 2026.

"Planning and Agency". Southern Society for Philosophy and Psychology, February 2026.

"Inquiry as Foraging". Annual Interdisciplinary Conference 50, January 2026.

"Moving Beyond Mappings". American Philosophical Association, Eastern Division, January 2026.

"Neuron Doctrines".* University of Pittsburgh, November 2025.

Panel Discussant.* Workshop on the Neuron Doctrine. University of Pittsburgh, October 2025.

"Neuron Doctrines". International Society for the Philosophy of the Sciences of the Mind, October 2025.

"A new folk psychological eliminativism". Philosophy of Neuroscience in the Gulf VIII, September 2025.

"Neuron Doctrines".* Purdue University, August 2025.

“Model Networks”. Society for Philosophy and Psychology, June 2025.
“Model Networks”. Scientific Understanding and Representation, May 2025.
“Model Networks”. Society for Philosophy and Neuroscience, May 2025.
“Model Networks”. Southern Society for Philosophy and Psychology, April 2025.
“Making Progress on Neural Representation”. American Philosophical Association, Pacific Division, April 2025.
“Inquiry, Search, and Foraging”.* Columbia University, March 2025.
“Model Networks”. Technical University Berlin, February 2025.
“Inquiry, Search, and Foraging”.* University of California – Irvine, February 2025.
“Model Networks”.* University of Notre Dame, January 2025.
“Foraging, Inquiry, and Thought: A Biologically Informed Theory of the Mind”.* Konrad Lorenz Institute, Vienna, Austria, October 2024.
“Foraging, Inquiry, and Thought: A Biologically Informed Theory of the Mind”. Macquarie University, NSW, Australia, October 2024.
“Composition in neural population dynamics”. Philosophy of Neuroscience in the Gulf VII, October 2024.
“Making Progress on Neural Representation”. Sofia University, Sofia, Bulgaria, September 2024.
“Scientific Inquiry as Foraging”.* Lingnan University, Hong Kong, September 2024.
“Making Progress on Neural Representation: Moving from Evidence to Content”. Nencki Open School, Warsaw, Poland, July 2024.
“Structure-Function Inference in Cognitive Neuroscience”. ESPP 31, July 2024.
“Bounds of Cognition”. University of Antwerp, May 2024.
“Context and Neural Function”. Southern Society for Philosophy and Psychology, March 2024.
“Scientific Inquiry as Foraging”.* University of Pittsburgh, February 2024.
“Neural Vehicles”. International Society for the Philosophy of the Sciences of the Mind, November 2023.
“Cognition Penetrates Vision”. Philosophy of Science Association Around the World, November 2023.
“Streams of Thought”.* University of Antwerp, October 2023.
“A Neurodynamical Alphabet”. Philosophy of Neuroscience in the Gulf VI, September 2023.
“Information Harvesting: Reasoning as Foraging in the Space of Propositions”. The Inquiry Network, April 2023.
“Computation with Neural Manifolds”.* University of California – Merced, April 2023.
“Inference is a Mental Process”. Southern Society for Philosophy and Psychology, March 2023.
“Computation with Neural Manifolds”. Philosophy of Science Association, November 2022.
“The New Central Executive”. Neurocognitive Foundations of Mind, November 2022.
“The New Central Executive”. Philosophy of Neuroscience in the Gulf V, September 2022.
“The New Central Executive”. University of Colorado – Boulder, September 2022.
“Computation with Neural Manifolds”. British Society for the Philosophy of Science, July 2021.

"Extended Intelligence". Kinds of Minds 2020 (postponed), June 2021.
"Computation with Neural Manifolds".* Neural Mechanisms Online, January 2021.
"Represent Represent". Philosophy of Neuroscience in the Gulf III, October 2020.
"On the Purported Impenetrability of Perception by Cognition".† Inter-American Congress of Philosophy, October 2019.
"Stream of Thought". American Philosophical Association, Central Division, February 2019.
"Streams of Thought: Search, Selection, and Content Preservation in Reasoning". Duke University, May 2018.
"The Explanatory Impotence of Deep Neural Networks". Southern Society for Philosophy and Psychology, March 2018.
"From Moth to Machine". American Philosophical Association, Central Division, February 2018.
"Mental Machines". American Philosophical Association, Eastern Division, January 2018.
"Levels and Layers". European Philosophy of Science Association, September 2017.
"Information Harvesting: Reasoning as Foraging in the Space of Inferences". American Philosophical Association, Eastern Division, January 2017.
"Dynamical Mechanisms and Models in Cognitive Neuroscience". Models and Simulations VI, University of Notre Dame, May 2014.

<u>Neuroscience</u>

"Navigation, Search, and the Algorithmic Basis of Cognition". University of Missouri, October 2026.
"Foraging, Inquiry, and Cognitive Architecture".* International Conference on Motivational and Cognitive Control, Helsinki, August 2026.
"Foraging, Inquiry, and Cognitive Architecture".* University of Birmingham, November, 2025.
"Foraging, Inquiry, and Cognitive Architecture". Carnegie Mellon University, October, 2025.
"Inquiry as Foraging".* Inmed, Marseilles, France, July 2024.
"Inquiry as Foraging".* Rochester Institute of Technology, February 2024.
"Inquiry as Foraging".* Georgia State University, January 2024.
"Inquiry as Foraging".* Emory University, August 2023.
"Inquiry as Foraging".* Queen's University, August 2023.
"The Quest for Information: Foraging as a Scaffold for Learning and Reasoning".* University of Miami, February 2023.
"The Quest for Information: Foraging as a Scaffold for Learning and Reasoning".* University of Wisconsin – Madison, February 2023.
"Primate Orbitofrontal Cortex Neurons Reflect Learning of States of Complex Environments". Icahn School of Medicine at Mount Sinai, MSNseminar, January 2023.
"The Quest for Information: Foraging as a Scaffold for Learning and Reasoning". University of Chicago, January 2023.
"The Quest for Information: Foraging as a Scaffold for Learning and Reasoning". Carnegie Mellon University, January 2023.

"Assessing Preferences for Different Foods in Nonhuman Primates during Unconstrained Behavior".† Society for Neuroscience 52, November 2022.
"What Is Foraging?"* Future of Foraging Seminar Series, May 2022, https://www.youtube.com/watch?v=vYQs1EgWYv0.
"Signatures of information foraging during learning of complex environments".* Innovators in Cognitive Neuroscience, October 2021, https://www.youtube.com/watch?v=LejOgvLQbCU.
"Two Views on the Cognitive Brain".* Ruhr University Bochum, September 2021.
"Two Views on the Cognitive Brain".* Duke University, July 2021.
"Two Views on the Cognitive Brain".* Brown University, July 2021.
"Information Seeking and Learning in Complex Environments".* Princeton University, September 2020.
"From Ecology to Neurobiology and Back: Foraging in Both the External and Internal Worlds".† COSYNE 2020.
"Information Seeking and Learning in Complex Environments".* Claremont Colleges, February 2020.
"Primates seek information to learn states of complex environments". Society for Neuroscience 49, October 2019.
"Posterior Cingulate Cortex and Adaptive Foraging". Society for Neuroscience 46, November 2016.
"From Bits to Policies: The Posterior Cingulate Encodes Environmental Information". Max Planck Institute for Intelligent Systems, July 2015.
"Adaptive Action and the Posterior Cingulate".* Princeton University, December 2014.
"Exploiting the Posterior Cingulate".* Princeton University, July 2014.
"Strategic Decision Making and the Cingulate Cortex". University of Pennsylvania, December 2012.

*: invited; †: accepted, not delivered.

Posters

Barack DL. 2024. Philosophy of Science Association.
Barack DL. 2024. Society for Philosophy and Psychology.
Barack DL and Salzman CD. 2023. COSYNE 20.
Barack DL and Salzman CD. 2022. CCN 6.
Barack DL and Jaegle A. 2022. Society for Philosophy and Psychology / European Society for Philosophy and Psychology Joint Meeting.
Barack DL and Salzman CD. 2018. Cognitive Computational Neuroscience (CCN) 2.
Barack DL and Salzman CD. 2017. Society for Neuroscience 47.
Bromberg-Martin EM, Platt ML, and Barack DL. 2016. COSYNE 13.
Barack DL, Li Y, Nassar MR, and Gold JI. 2015. Society for Neuroscience 45.
Barack DL. 2015. Society for Philosophy and Psychology.
Barack DL, Gariépy JF, and Platt ML. 2014. Society for Neuroscience 44.
Barack DL, Gariépy JF, and Platt ML. 2014. COSYNE 11.
Barack DL, Gariépy JF, and Platt ML. 2013. Society for Neuroscience 43.
Barack DL and Platt ML. 2013. COSYNE 10.
Barack DL, Hayden BY, and Platt ML. 2012. Society for Neuroscience 42.

Barack DL, Hayden BY, Pearson JM, and Platt ML. 2012. Computational and Systems Neuroscience (COSYNE) 9.
Barack DL, Hayden BY, Pearson JM, and Platt ML. 2011. Society for Neuroscience 41.
Harris J, Barack DL, and Woldorff MW. 2010. Society for Neuroscience 40.
Barack DL and Woldorff MW. 2009. Society for Neuroscience 39.

Teaching

Carnegie Mellon University, Pittsburgh, PA August 2026-December 2026
Lecturer, "Meaning and Doing: Philosophy of Mind I"
Pitzer College, Claremont, CA January 2025-May 2025
Lecturer, "Introduction to Philosophy of Science"
Lecturer, "Topics in Philosophy of Neuroscience"
Lingnan University, Hong Kong September 2024-December 2024
Lecturer, "Critical Thinking"
Duke University, Durham, NC January 2013-May 2014
Teaching Assistant, "Neuroethics", Philosophy Department
Lecturer, "Introduction to Philosophy" and "Seminar in Theories of Cognition", Philosophy Department
University of Wisconsin-Milwaukee, Milwaukee, WI August 2005-May 2007
Teaching Assistant and Lecturer, "Introduction to Philosophy", Philosophy Department
Pomona College, Claremont, CA September 2003-December 2003
Research and Teaching Assistant, "Consciousness and Cognition", Psychology Department

Service

<u>Societies</u>
Society for Philosophy and Neuroscience
- Founding Member
- Chair (2024 -)

<u>Ad Hoc Referee</u>

Journals
Synthese, Ergo, British Journal for the Philosophy of Science, Australasian Journal of Philosophy, Nature Neuroscience, Nature Reviews Neuroscience, Neuron, International Studies in the History and Philosophy of Science, Current Biology, European Journal for the Philosophy of Science, Topics in Cognitive Science, Trends in Cognitive Science, Biology and Philosophy

Books
Oxford University Press, MIT Press

References

Philosophy:

Dr. Bence Nanay
Professor of Philosophy and BOF Research Professor
Centre for Philosophical Psychology
University of Antwerp
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Dr. Christopher A B Peacocke
Johnsonian Professor of Philosophy
Columbia University
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Dr. John Morrison
Associate Professor of Philosophy
Barnard College, Columbia University
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Dr. Michael Weisberg
Professor and Chair of Philosophy
University of Pennsylvania
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Dr. Alex Rosenberg
Chair and R. Taylor Cole Professor of Philosophy
Duke University
alexrose@duke.edu

Dr. Walter Sinnott-Armstrong
Chauncey Stillman Professor of Ethics and Professor of Kenan Institute for Ethics Duke University
ws66@duke.edu

Dr. Felipe De Brigard
Fuchsberg-Levine Family Associate Professor of Philosophy
Duke University
felipe.debrigard@duke.edu

Neuroscience:

Dr. Michael L Platt
James S. Riepe University Professor
Professor of Neuroscience, Professor of Psychology, and Professor of Marketing
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